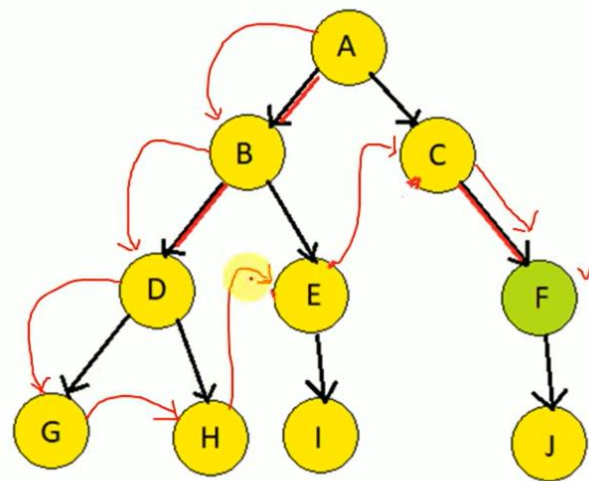
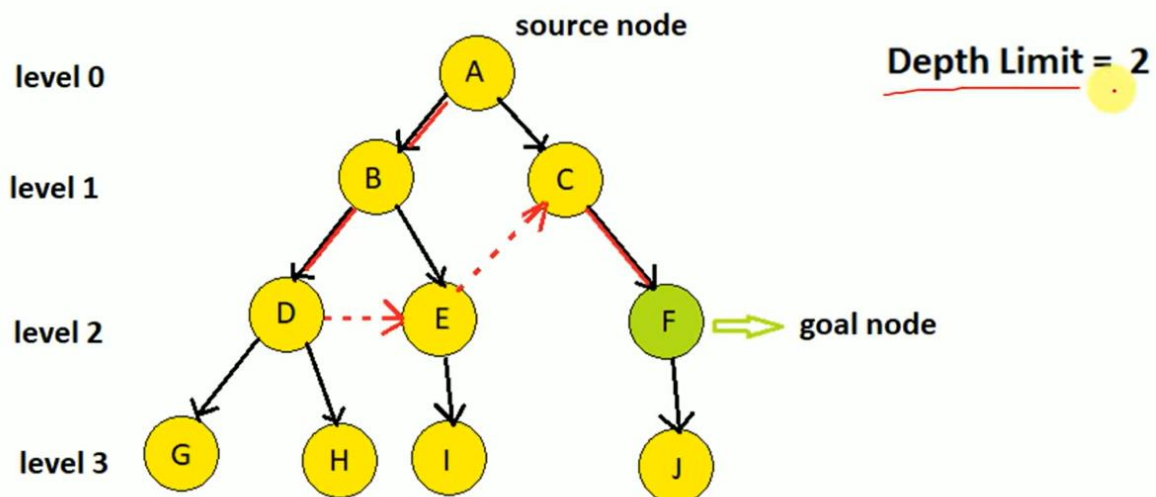


Depth First Search Algorithm in Artificial Intelligence

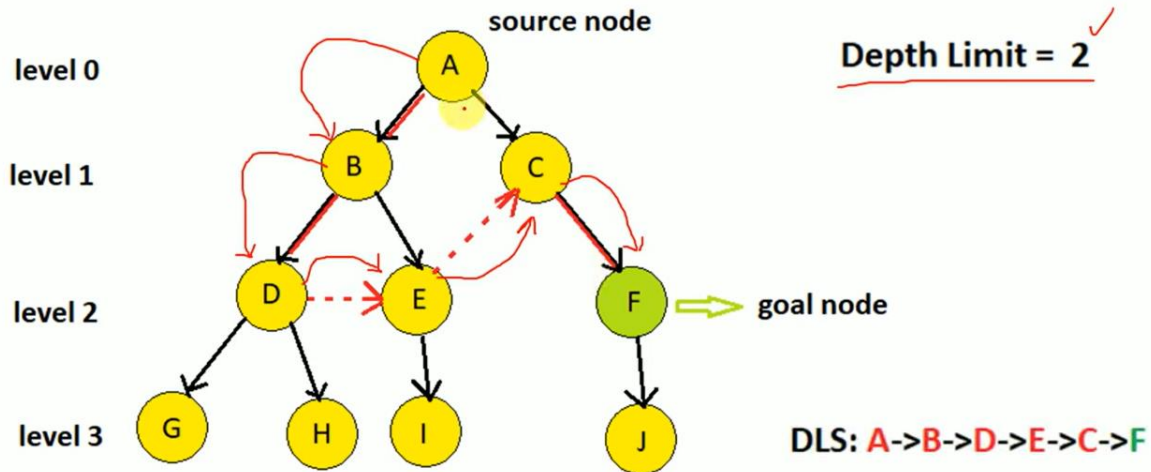


DFS: A→B→D→G→H→E→I→C→F

Depth Limited Search Algorithm in Artificial Intelligence



Depth Limited Search Algorithm in Artificial Intelligence



Hill Climbing Search Algorithm in Artificial Intelligence

- In hill climbing the basic idea is to always head towards a state which is better than the current one.
- So, if you are at town A and you can get to town B and town C (and your target is town D) then you should make a move IF town B or C appear nearer to town D than town A does.

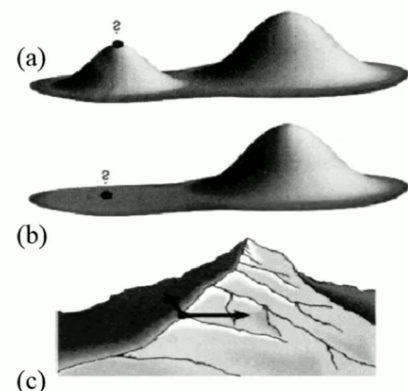
Hill Climbing Search Algorithm in Artificial Intelligence

1. Evaluate the initial state. If it is also goal state then return it, otherwise continue with the initial state as the current state.
2. Loop until the solution is found or until there are no new operators to be applied in the current state
 - a) Select an operator that has not yet been applied to the current state and apply it to produce new state
 - b) Evaluate the new state
 - i. If it is a goal state, then return it and quit
 - ii. If it is not a goal state but it is better than the current state, then make it as current state
 - iii. If it is not better than the current state, then continue in loop.

Hill Climbing Search Algorithm - Drawbacks

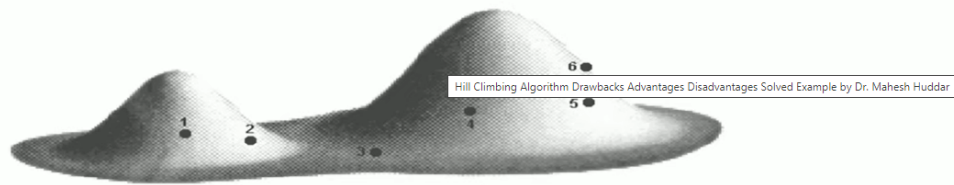
This simple Hill Climbing policy has three well-known drawbacks:

1. **Local Maxima**: a local maximum as opposed to global maximum.
2. **Plateaus**: An area of the search space where evaluation function is flat, thus requiring random walk.
3. **Ridge**: Where there are steep slopes, and the search direction is not towards the top but towards the side.



Hill Climbing Search Algorithm - Drawbacks

- In each of the previous cases (local maxima, plateaus & ridge), the algorithm reaches a point at which no progress is being made.
- A solution is to do a **random-restart hill-climbing** - where random initial states are generated, running each until it halts or makes no discernible progress. The best result is then chosen.



Hill Climbing: Disadvantages

- Hill climbing using **local information** :
Decides what to do next by looking only at the “immediate” consequences of its choices.
- Will terminate when at local optimum.
- The order of application of operators can make a big difference.
- **Global information** might be encoded in heuristic functions.

Hill Climbing Solved Example

Init State



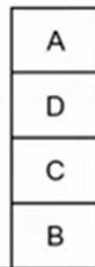
Goal State



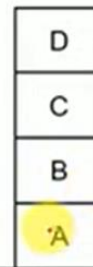
Local and
Global Heuristic
Function
Artificial
Intelligence

Hill Climbing: Local Heuristic function

Start

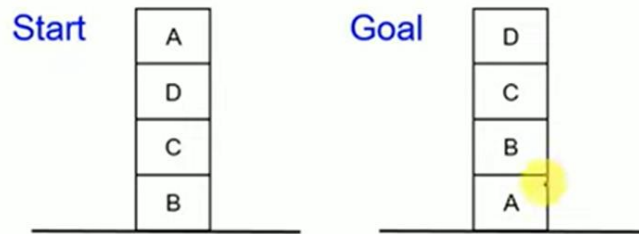


Goal



Blocks World

Hill Climbing: Local Heuristic function



Blocks World

Local heuristic:

+1 for each block that is resting on the thing it is supposed to be resting on.

-1 for each block that is resting on a wrong thing.

Hill Climbing: Local Heuristic function



Blocks World

Local heuristic:

+1 for each block that is resting on the thing it is supposed to be resting on.

-1 for each block that is resting on a wrong thing.

Hill Climbing: Local Heuristic function



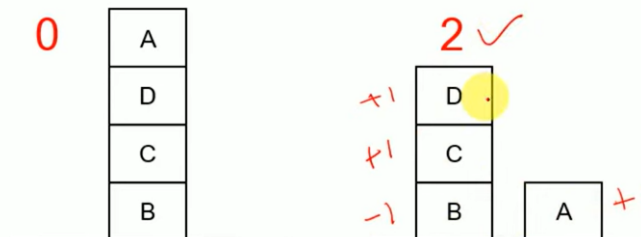
Blocks World

Local heuristic:

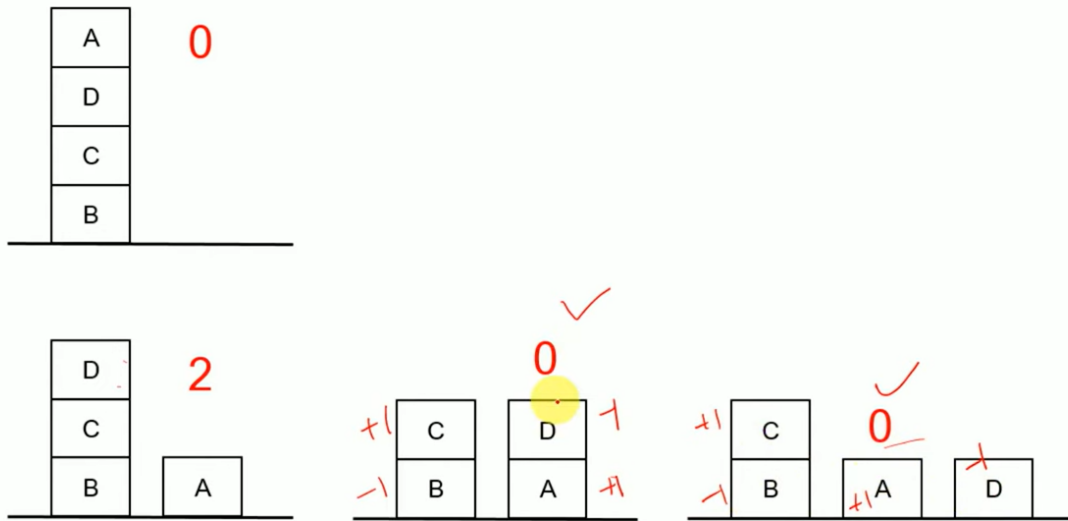
+1 for each block that is resting on the thing it is supposed to be resting on.

-1 for each block that is resting on a wrong thing.

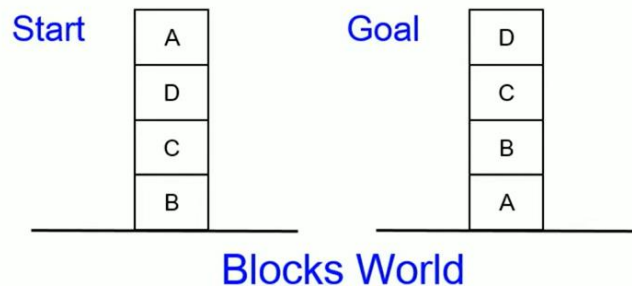
Hill Climbing: Local Heuristic function



Hill Climbing: Local Heuristic function



Hill Climbing: Global Heuristic function

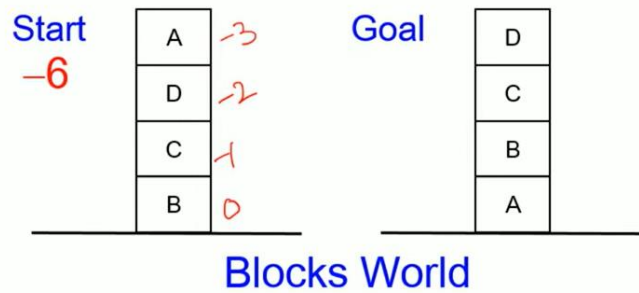


Global heuristic:

For each block that has the correct support structure: **+1** to every block in the support structure.

For each block that has a wrong support structure: **-1** to every block in the support structure.

Hill Climbing: Global Heuristic function

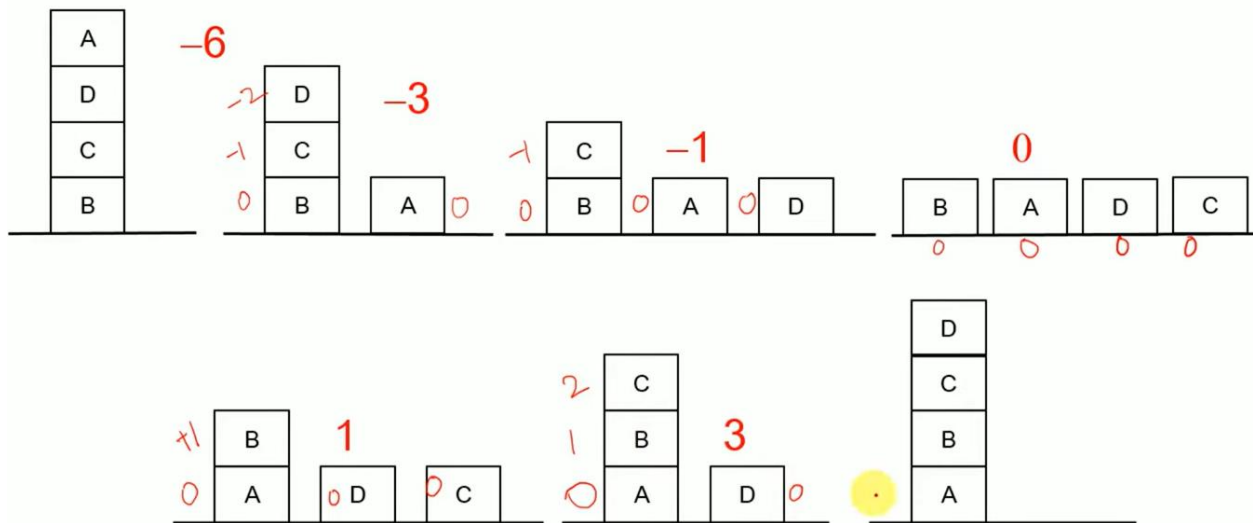


Global heuristic:

For each block that has the correct support structure: **+1** to every block in the support structure.

For each block that has a wrong support structure: **-1** to every block in the support structure.

Hill Climbing: Global Heuristic function



Hill Climbing: Global Heuristic function

